

Figure 1. The original lead II plot of one of the 12-leads given for a patient experiencing AFib.

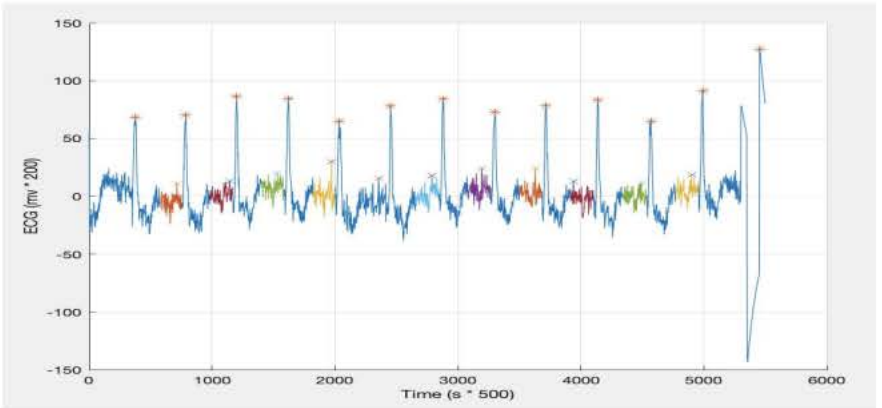


Figure 2. Application of a bandpass filter, detrending the data, and finding the R and P waves of each complex.

```
>> SystemsFinalCode
Consistent R intervals.
WARNING: Missing P wave. Seek medical attention.
```

Figure 3. The output of the code. It shows that all R intervals are consistent enough to not cause alarm, but the P intervals are inconsistent because of fibrillatory waves in between QRS complexes. This results in a warning message.

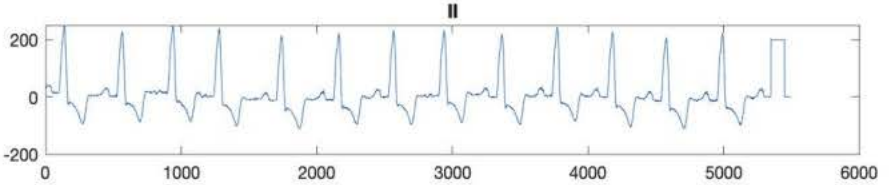


Figure 4. The original lead II plot of one of the 12-leads given for a patient not experiencing AFib.

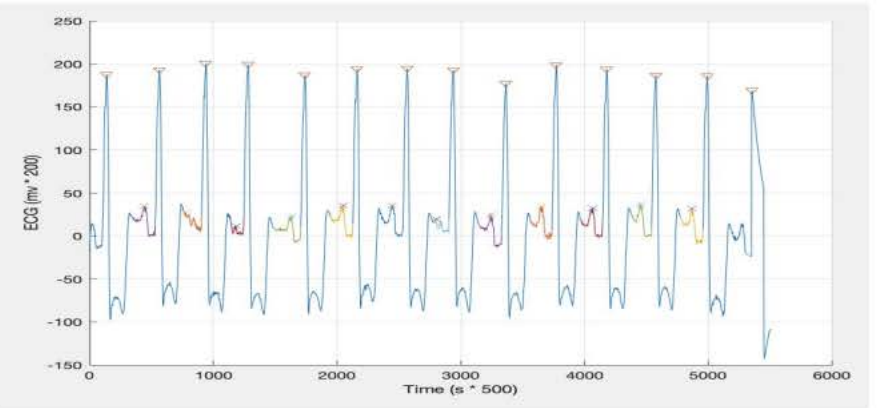


Figure 5. Application of a bandpass filter, detrending the data, and finding the R and P waves of each complex.

```
>> SystemsFinalCode
Consistent R intervals.
Consistent P waves.
```

Figure 6. Both R and P waves with consistent intervals; therefore, no warning messages occur.